

Tor Vergata University of Rome



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

FACULTY OF ENGINEERING

**COMPUTER SCIENCE, CONTROL AND GEOINFORMATION
CYCLE XXXVII**

PhD Thesis

TITLE

ADVISOR

Prof. Sergio Galeani

CANDIDATE

Roberto Masocco

COORDINATOR

Prof. Francesco Quaglia

A.Y. 2023/2024



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

COMPUTER SCIENCE, CONTROL AND GEOINFORMATION
CYCLE XXXVII

TITLE

PHD THESIS

Advisor:

Prof. Sergio Galeani

Signature:_____

Coordinator:

Prof. Francesco Quaglia

Signature:_____

Candidate:

Roberto Masocco

Signature:_____

A.Y. 2023/2024

Tor Vergata University of Rome, Faculty of Engineering
Via del Politecnico 1, Rome, Italy

Wings are a constraint that makes
it possible to fly.
— Robert Bringhurst

To my parents...

Acknowledgements

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Lausanne, 12 Mars 2011

D. K.

Preface

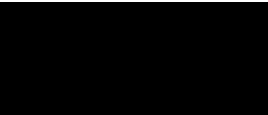
A preface is not mandatory. It would typically be written by some other person (eg your thesis director).

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

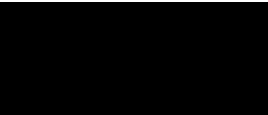
Lausanne, 12 Mars 2011

T. D.

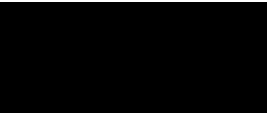


Contents

Acknowledgements	i
Preface	iii
List of figures	vii
List of tables	ix
1 Introduction	1
Introduction	1
A An appendix	3
Bibliography	5



List of Figures



List of Tables

1 Introduction

Pervasiveness. A word that, as its meaning almost suggests, is gradually making its way into everyday narratives, effectively describing the impact of new technologies on our daily lives. It is evident how many tasks and challenges we now rely on, in whole or part, to machines: a growing number that keeps increasing. As we become able to accomplish new tasks and solve more problems with the help of automated devices, new scenarios and opportunities emerge. Research, medicine, logistics, and entertainment: these are just a few of the countless fields in which the introduction of digital and autonomous devices—ever more sophisticated over the decades—has led to innovations at an unprecedented pace. However, this ever-growing diffusion comes with an increase in complexity, both in design and implementation, something that is often praised but more quietly acknowledged as a source of new challenges and difficulties. This is, above all, due to the multitude and variety of tasks that these devices should now perform.

Footnote should be cited here Galeani, Masocco, and Sassano 2022; Galperti et al. 2024.

A An appendix

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Bibliography

- Galeani, S., R. Masocco, and M. Sassano (Dec. 2022). “Dynamic control allocation for sampled-data closed-loop systems”. In: *Proc. of the IEEE 61st Conference on Decision and Control (CDC)*. IEEE, pp. 4206–4211.
- Galperti, C. et al. (Nov. 2024). “Overview of the TCV digital real-time plasma control system and its applications”. In: *Fusion Engineering and Design* 208, p. 114640.